

M.TECH SPECIALIZATION SYLLABI BOOKLETS

Detailed Course Outlines for AIML, AMT, TE, and EAD Specialisations

Ref: RGUKT/PG/SYLLABUS-2024

Effective: 2024-2026 Curriculum

Authority: PG Board of Studies, RGUKT-AP

1. 1. Artificial Intelligence & Machine Learning (CSE)

- **Core Subjects:** Mathematical Foundations of Data Science, Deep Learning Architectures, Natural Language Processing, Computer Vision, Large Scale Distributed AI.
- **Tools & Platforms:** PyTorch, TensorFlow, CUDA GPU Clusters, Hugging Face, Ray Distributed Framework.

2. 2. Advanced Materials Technology (MME)

- **Core Subjects:** Advanced Thermodynamics & Kinetics, Nanostructured Materials, Microscopic Characterization (SEM/TEM/XRD), Computational Materials Modeling (DFT).
- **Tools & Labs:** VASP, Thermo-Calc, High-Resolution Optical and Electron Microscopy.

3. 3. Transportation Engineering (Civil)

- **Core Subjects:** Traffic Flow Theory, Pavement Analysis & Sustainable Materials, Intelligent Transportation Systems (ITS), Urban Transit Logistics.
- **Tools & Labs:** VISSIM, ArcGIS, KENPAVE, Road Safety Audit Systems.

4. 4. Engineering Analysis & Design (Mechanical)

- **Core Subjects:** Advanced Finite Element Analysis (FEA), Computational Fluid Dynamics (CFD), Mechanics of Composites, Vibrations & Structural Reliability.
- **Tools & Labs:** ANSYS Fluent & Mechanical, Abaqus, HyperMesh, OpenFOAM.